

Week 10 - Multi-Agent Systems using LangGraph

Questions

Q1. What is the primary purpose of a Multi-Agent system in LangGraph?

- A. To train multiple LLMs simultaneously
- B. To divide a complex task among specialized agents
- C. To store embeddings
- D. To reduce vector database size

Q2. In a Research–Writer workflow, what is the responsibility of the Research Agent?

- A. Generate the final report
- B. Retrieve and gather relevant information
- C. Deploy the application
- D. Validate API keys

Q3. Which agent is responsible for converting research findings into a well-structured document?

- A. Retriever Agent
- B. Writer Agent
- C. Router Agent
- D. Memory Agent

Q4. Why is LangGraph suitable for implementing Multi-Agent workflows?

- A. It supports graph-based execution of multiple interconnected agents
- B. It automatically trains language models
- C. It replaces vector databases
- D. It eliminates the need for prompts

Q5. Which component is commonly shared between multiple agents in a LangGraph workflow?

- A. Docker Container
- B. State
- C. GPU Memory
- D. Python Environment

Q6. In a Research–Writer workflow, what information is typically passed from the Research Agent to the Writer Agent?

- A. API keys
- B. Retrieved research or context
- C. Source code files
- D. Environment variables

Q7. Which of the following best describes the execution order in a simple Research–Writer graph?

- A. Writer → Research → END
- B. Research → Writer → END
- C. END → Writer → Research
- D. Writer → END → Research

Q8. What is the main advantage of separating Research and Writer Agents?

- A. Reduces internet speed
- B. Improves modularity and allows each agent to specialize in its task
- C. Eliminates the need for prompts
- D. Increases embedding dimensions

Q9. In LangGraph, what determines which agent executes next?

- A. Prompt Length
- B. Edges in the Graph
- C. Temperature
- D. Token Count

Q10. Which of the following best describes the State in a Multi-Agent LangGraph application?

- A. It stores only API credentials
- B. It acts as shared memory between agents during execution
- C. It contains only the final response
- D. It stores Python packages